INTERVIEW-PREP BOOKLET · NACF INBOUND FLUENCY · BUSINESS-FIRST, SYSTEMS AS ENABLERS

NACF Inbound — how capacity and inventory are managed

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6) · interview loop October 29–30, 2026

The panel this booklet serves

1 · Snigdha Tripathy — Sr Manager, Supply Chain — NACF · **HIRING MANAGER** · ex-EU IXD / placement leader ($80MM savings 2021; built the IXD team ground-up, +$50MM 2020; 50+ FC launches Amazon.in; Kaizen/Lean trainer; 11+ yrs Amazon)

2 · Ryan Hoerger — Manager, Placement / Transfer Analytics · ex-Sr BIE · Chicago Booth MBA, Siebel Scholar · Nike analytics · statistical rigor

3 · Rocky Goforth — Sr Manager, Infrastructure

4 · Zhaowen Lan — Sr Manager, Supply Chain — peer

5 · Venkata Josyula — Principal PM, FBA · bar raiser

> **Strategy in one line.** Sound like someone who has run inbound — not someone who memorized a glossary. Every mechanism below is spoken as a business process with a constraint, a metric and a lever; every number is worked end-to-end so the arithmetic can be done out loud in the room. Inbound in one breath: *the right inventory, reserved into the right node, received and stowed inside capacity, made available where the demand is — with the plans traveling ahead of the freight and the variance closing the loop.*

Honesty labels, on every mechanism: **[well-known public Amazon practice]** — published, talkable-in-public Amazon mechanisms (AWD, IXD, appointment scheduling, random stow, IPI components, WBR) · **[industry standard]** — textbook supply chain (UPH labor planning, z·σ·√LT safety stock, S&OP, queueing approximations) · **[illustrative]** — numbers invented to make the math legible. No confidential Amazon internals are asserted anywhere; no number is claimed as Amazon's own.

How to use. §1–§4 are the fluency: read the diagrams until you can redraw them, then recompute every green math block until the substitutions feel like your own. §5 maps your master-packet stories onto inbound concepts. §6 is per-interviewer: what each of the five will likely ask, and the questions to ask them. §7 is the math appendix — every formula symbolic → substituted → computed, for the night before.

Canon: amazon/amazon-l6-scm-master-packet.html (stories S2, S6, S8, S9, S11, S12, S13, S18, S19 with their basis labels) · panel packets 1–5 (per-interviewer lenses). Sibling booklet file; the five panel packets are untouched by this document.

The 60-second answer — "what is NACF inbound?"

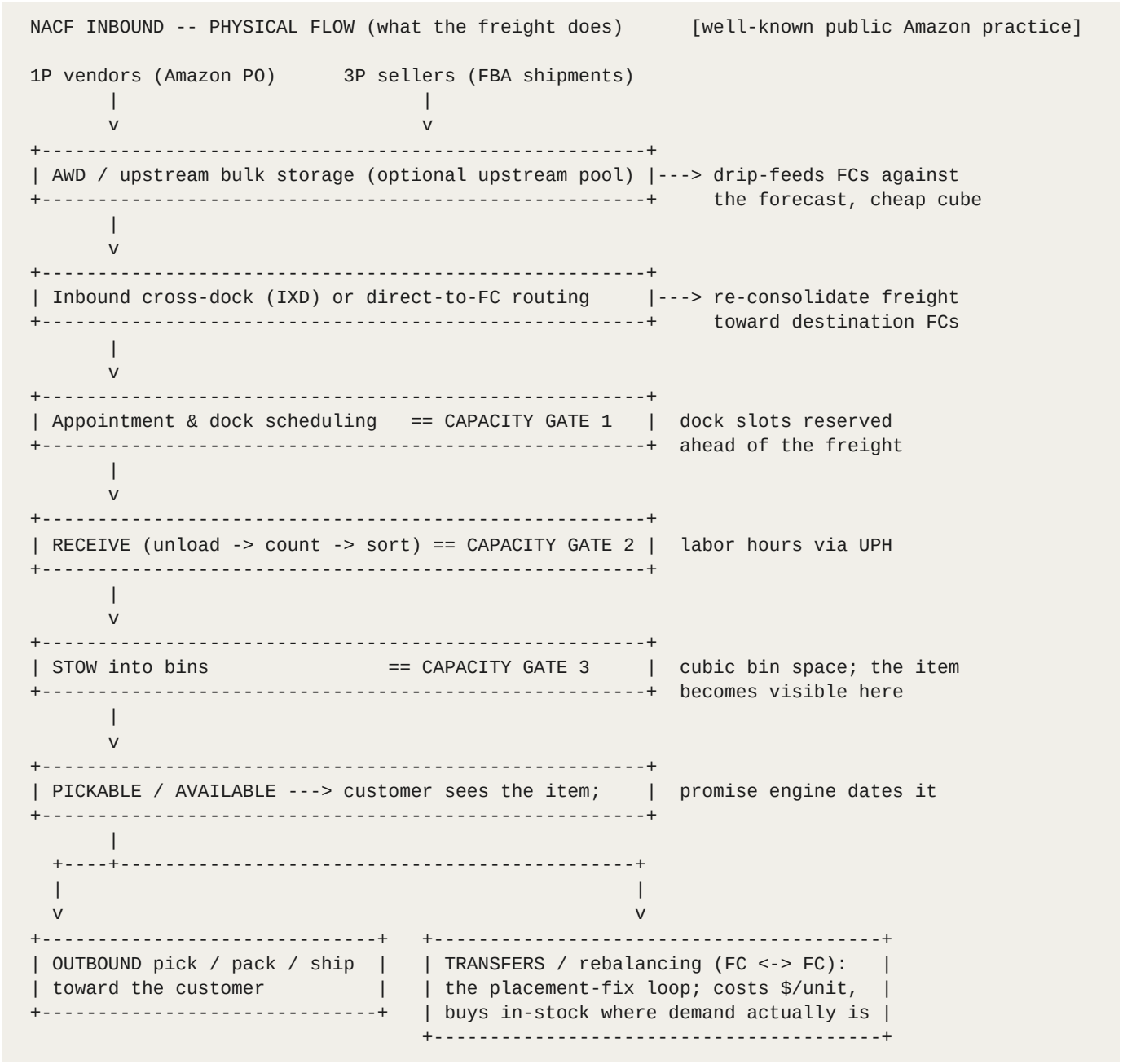
> **SAY THIS VERSION OUT LOUD, THEN STOP**
>
> "Inbound is everything from the vendor or seller's dock to the moment an item is pickable in front of a customer promise — and it is managed as two coupled problems. **Capacity**: three physical currencies — dock-appointment slots, labor hours, and cubic storage — planned from the demand forecast through a receive plan and a UPH-based labor plan, with peak buffers sized off the forecast distribution, not the average. **Inventory**: where the units land and how long they last — placement by distance-to-customer against cost, weeks of cover by SKU, safety stock positioned where the variability actually is, and a transfer loop that rebalances what placement got wrong. The information flow travels ahead of the freight: forecast → inbound plan → placement decision → capacity reservation, and the actuals come back through the WBR as variance. My job in a seat like this is to keep that loop honest — measured lead times, not brochure ones; buffers released on evidence; capacity added against the constraint that actually binds."

Everything after this page unpacks that paragraph. The discipline that makes it credible: every mechanism carries its label, and every quantity in the worked examples can be recomputed from the row above it — do the arithmetic out loud in the interview and the fluency reads as experience.

> **The one mental model to keep on top.** Inbound is a *reservation system* feeding a *positioning system*: you reserve capacity (slots, hours, cube) against a plan, and you position inventory (placement, cover, buffers) against demand. Most inbound failures are one system lying to the other — freight arrives that capacity never reserved, or inventory sits where demand isn't. The metrics in §2–§3 exist to catch exactly those two lies early.

1 · The inbound machine, end-to-end

The physical flow first — what the freight does. Each node is labeled with the constraint that binds it; the three capacity gates are the whole of §2. **[well-known public Amazon practice]** for the node names (AWD, IXD, appointment scheduling, random-stow-style bin visibility); the diagram itself is **[illustrative]**.



NODE	WHAT IT IS (BUSINESS LANGUAGE)	WHAT CONSTRAINS IT	THE METRIC THAT WATCHES IT	LABEL
Vendor / seller ship	1P purchase orders and 3P FBA shipments; the plan's first promise is the ship date and the ASN (advance shipment notice).	Vendor lead time and OTIF (on-time-in-full); ASN data quality.	OTIF %, fill rate, ASN accuracy.	**[industry standard]**

NODE	WHAT IT IS (BUSINESS LANGUAGE)	WHAT CONSTRAINS IT	THE METRIC THAT WATCHES IT	LABEL
AWD / upstream	Low-cost bulk storage upstream of the FC network; holds inventory cheap and replenishes the network as the forecast firms.	Upstream cube; replenishment lead time into FCs.	Upstream weeks of cover; replenishment dwell.	**[well-known public Amazon practice]**
IXD / direct FC	Inbound cross-dock sorts and re-consolidates inbound freight toward destination FCs; some freight books direct.	Cross-dock throughput; linehaul schedule.	IXD dwell time; trailer utilization.	**[well-known public Amazon practice]**
Appointment / dock	Carriers book delivery appointments; the dock door is the first scarce, schedulable resource in the building.	Dock doors × receiving hours × unload rate.	Dock utilization %, trailer dwell, missed-appointment count.	**[well-known public Amazon practice]**
Receive	Unload, count, inspect, sort to stowable state; where physical freight becomes system inventory.	Labor hours (units per hour, UPH) and conveyor/sort rates.	Receive UPH, backlog in trailers/cartons, receive accuracy.	**[industry standard]**
Stow	Items into bins; the item becomes findable and sellable — availability starts here, not at the dock.	Cubic bin space and stow labor.	Stow UPH, bin utilization %, time-to-available.	**[well-known public Amazon practice]**
Pickable / available	Inventory the promise engine can sell against; the boundary between inbound and outbound.	Nothing inbound — this is the finish line.	In-stock rate; time from dock to available.	**[industry standard]**
Transfers / rebalancing	FC-to-FC moves that fix placement error after the fact; the network's insurance policy, priced per unit.	Transfer cost vs stockout cost (the breakeven in §3).	Transfer units/wk, transfer cost/unit, avoided stockouts.	**[well-known public Amazon practice]**

The information flow — plans travel ahead of the freight

Every physical move above is preceded by an information move; when the two drift apart, the WBR is where it gets caught. **[industry standard]** structure; Amazon vocabulary (WBR, promise engine) **[well-known public Amazon practice]**.

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INFORMATION FLOW -- plans travel ahead of the freight     [industry standard]

demand forecast (SKU x region x week)
       |
       v
inbound plan: what to buy / book, and when it must land  <---> vendor & seller
       |                                                       lead times, OTIF
       v
placement decision: WHICH node receives it --
  distance-to-customer vs cost vs node capacity      <--- the hiring manager's
       |                                                  home turf (EU IXD placement)
       v
capacity reservation: dock slots + labor plan (UPH -> hours) + cube
       |
       v
receive schedule -> stow plan -> availability signal
       |
       v
promise engine / search in-stock ---> the customer sees a delivery date
       |
       v
actuals vs plan --> variance --> next WBR input (the loop that keeps it honest)
```

> **How to sound like you've run it.** Name the seam, not the node. "The dock" is trivia; "the seam where the inbound plan meets the appointment calendar — and what happens when a vendor ships three days early into a full dock" is experience. Each worked example in §2–§3 is one of those seams, with the arithmetic attached.

2 · Capacity management — the three currencies and the planning loop

Inbound capacity is three currencies that convert into each other only through the plan. **[industry standard]** framework; all numbers **[illustrative]**.

CAPACITY TYPE	UNIT OF ACCOUNT	WHAT DRIVES CONSUMPTION	HOW YOU PLAN IT	WHEN IT BINDS
Dock / appointment slots	Trailer-unloads per day = doors × hours × unload rate	Truckload arrivals; vendor/carrier appointment discipline; trailer mix (floor-loaded vs palletized).	Receive plan → trailers/day → doors at an unload rate, capped at ~85% utilization (worked in CAP-A, CAP-D).	First, at peak — it is the least elastic currency (you cannot stack trailers at a door).
Labor hours	Hours = units ÷ UPH, grossed up for indirect work and learning curves	Units received/stowed; associate proficiency; process (conveyable vs hand-stack).	UPH standards → direct hours → paid hours → headcount by shift (CAP-B).	Second — elastic but slow: hiring and the learning curve lag the plan by weeks.
Cubic storage	Cubic feet = on-hand units × cube per unit	Inventory target (days of cover), unit cube, bin geometry / honeycomb loss.	Inventory plan → cube needed vs effective bin capacity (CAP-C).	Last, then suddenly — it looks fine at 49% and is a wall at 90%; the least flexible currency mid-peak.

The planning loop — forecast to floor

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CAPACITY PLANNING LOOP                                          [industry standard]

demand forecast (units/day, with a distribution -- p50 ... p90)
       |
       v
receive plan: units/day -> cartons -> pallets -> trailers/day (CAP-A)
       |
       +---------------> dock reservation: trailers/hour vs doors at ~85% ceiling
       |
       v
labor plan: units / UPH = direct hours -> +indirect % -> /shift-hours = heads (CAP-B)
       |
       v
cube check: on-hand units x cube/unit vs effective bin capacity (CAP-C)
       |
       v
peak buffer: fixed capacity to ~p50-p70, flex (temp labor, OT, appointment
smoothing) for the tail (CAP-E) -- then WBR reads actuals vs all of it
```

The metrics that watch capacity

METRIC	DEFINITION / FORMULA	RED FLAG	LABEL
Dock utilization %	`utilization = scheduled unloads ÷ (doors × hours × unload rate)`	Sustained > ~85% — dwell goes nonlinear (CAP-D).	**[industry standard]**
Dwell time	`dwell = trailer arrival → door departure (hours)`	Rising dwell with flat volume = variability, not volume, is the problem.	**[industry standard]**
Receive backlog	`backlog_days = unreceived units ÷ receive capacity (units/day)`	> 0.5 day carried into the next shift; it compounds.	**[industry standard]**

METRIC	DEFINITION / FORMULA	RED FLAG	LABEL
Forecast variance	`variance = (actual − plan) ÷ plan, by week and by node`	Same-sign variance three weeks running = a bias, not noise; fix the plan, not the floor.	**[industry standard]**

CAP-A · REQUIRED RECEIVE CAPACITY — UNITS/DAY → TRAILERS → DOCK DOORS

One illustrative FC receiving 120,000 units/day. Every row recomputes from the rows above; say the chain out loud and you sound like the floor.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Inbound volume	`V = 120,000 units/day`	illustrative receive plan	120,000 units/day
2	Cartons per day	`120,000 ÷ 6 = 20,000`	6 units/carton (illustrative)	20,000 cartons/day
3	Pallets per day	`20,000 ÷ 40 = 500`	40 cartons/pallet (illustrative)	500 pallets/day
4	Truckloads per day	`500 ÷ 26 = 19.2`	26 pallets per 53-ft trailer [industry standard]	19.2 trailers/day
5	Trailer arrival rate	`19.2 ÷ 10 = 1.92`	10 receiving hours/day (illustrative)	1.92 trailers/hour
6	Raw doors required	`1.92 ÷ 0.5 = 3.84`	unload rate 0.5 trailer/door-hour = 2 h/trailer (illustrative)	3.84 doors
7	Doors at the 85% ceiling	`3.84 ÷ 0.85 = 4.5 → round up`	utilization ceiling ~85% [industry standard]	**5 dock doors planned**

The sentence to say: "Twelve trailers an hour isn't the plan — the plan is five doors, because at 85% utilization the doors stop being fungible and dwell starts compounding."

CAP-B · LABOR HOURS FROM UPH — WITH THE LEARNING CURVE PRICED IN

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Direct receive hours	`120,000 ÷ 300 = 400`	mature-associate 300 UPH (illustrative)	400 direct hours/day
2	Paid hours (indirect gross-up)	`400 × 1.15 = 460`	15% indirect allowance (illustrative)	460 paid hours/day
3	Headcount	`460 ÷ 10 = 46`	10-hour shifts (illustrative)	46 associates/day
4	Learning curve, week 1	`300 × 0.60 = 180 UPH`	new associate at 60% of standard (illustrative)	180 UPH effective
5	Blended UPH at 30% new	`0.7×300 + 0.3×180 = 210 + 54 = 264`	Q4 ramp: 30% of staff in week 1–2 (illustrative)	264 UPH blended
6	Direct hours at blended UPH	`120,000 ÷ 264 = 454.5`	same volume as step 1	454.5 h = **+54.5 h (+13.6%)**

The sentence to say: "The hiring plan isn't the headcount math — it's the learning-curve math. Thirty percent new hires isn't a 30% problem, it's a 13.6% tax on every received unit for two weeks, and the peak plan has to pay it before the first trailer arrives."

CAP-C · CUBIC CAPACITY — UNITS → CUBE → % OF THE BUILDING

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	On-hand target	`I = 2,000,000 units`	illustrative network-position target for this FC	2.0M units
2	Cube required	`2,000,000 × 0.25 = 500,000`	0.25 ft³/unit average (illustrative)	500,000 ft³
3	Nominal utilization	`500,000 ÷ 1,200,000 = 41.7%`	1.2M ft³ usable bin capacity (illustrative)	41.7% of nominal
4	Effective capacity (honeycomb)	`1,200,000 × 0.85 = 1,020,000`	15% partial-bin loss [industry standard]	1,020,000 ft³ effective
5	True utilization	`500,000 ÷ 1,020,000 = 49.0%`	against effective, not nominal	**49.0% — plan against this**
6	Days of cover check	`2,000,000 ÷ 120,000 = 16.7 days`	same 120K units/day flow as CAP-A	16.7 days
7	Surge consumption	`3 × 120,000 × 0.25 = 90,000 ft³; 90,000 ÷ 1,020,000 = 8.8%`	a 3-day inbound surge ahead of forecast	8.8% of effective cube in 3 days

The sentence to say: "Cube is the currency that lies — 42% nominal is 49% effective, and a three-day vendor early-ship surge eats nine points of the building before the WBR sees it."

CAP-D · WHY DWELL EXPLODES ABOVE ~85% DOCK UTILIZATION

Queueing approximation: with variability in arrivals and unload times, expected wait scales with ρ/(1−ρ) and total dwell with service ÷ (1−ρ), where ρ = utilization **[industry standard — Kingman-style approximation, qualitative use]**.

STEP	UTILIZATION P	QUEUE FACTOR P/(1−P)	DWELL AT 2.0 H SERVICE: 2 ÷ (1−P)	READING
1	70%	`0.70 ÷ 0.30 = 2.33`	`2.0 ÷ 0.30 = 6.7 h`	comfortable
2	85%	`0.85 ÷ 0.15 = 5.67`	`2.0 ÷ 0.15 = 13.3 h`	the knee — the ceiling in CAP-A
3	95%	`0.95 ÷ 0.05 = 19.0`	`2.0 ÷ 0.05 = 40.0 h`	a 2-hour unload becomes a 40-hour trailer stay
4	85% → 95%	`19.0 ÷ 5.67 = 3.4×`	`40.0 ÷ 13.3 = 3.0×`	**+10 points of utilization ≈ 3× the dwell**

The sentence to say: "Above about 85% the building stops failing linearly — the last ten points of utilization triple your dwell. So the plan holds dock slack the way inventory holds safety stock: you don't apologize for either until the week you needed it."

CAP-E · PEAK BUFFER SIZING — P50 VS P90

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Peak-week distribution	`p50 = 120,000; p90 = 156,000`	illustrative; p90/p50 = 1.30	+36,000 units/day at p90
2	Build to p50 — one p90 day	`36,000 ÷ 120,000 = 0.30 day`	shortfall carried as backlog	0.30 day of backlog
3	Build to p50 — three p90 days	`3 × 36,000 = 108,000; ÷ 120,000 = 0.9`	backlog compounds across the shift handoffs	~1 full day of backlog
4	Build to p90 — a p50 day	`36,000 ÷ 156,000 = 23%`	idle capacity on the median day	23% idle
5	Labor to flex the tail	`156,000 ÷ 264 = 591 h; 591 − 454.5 = 136 h; ÷ 10 ≈ 14`	blended 264 UPH from CAP-B; 10-h shifts	**≈14 flex associates**

The sentence to say: "Fixed capacity to the median, flex the tail. Building the building to p90 means paying 23% idle on the average day; building to p50 means a three-day p90 run is a full day of backlog — so the answer is temp labor, overtime and appointment smoothing, sized at about fourteen heads on this volume."

3 · Inventory management, inbound lens — where the units land and how long they last

Capacity decides how fast freight moves; inventory decisions decide whether the right units are in the right building. This is the hiring manager's home turf — placement was her EU IXD charter — so the placement math (INV-C) gets said slowly, with the constraint and the shadow price named.

Placement strategy — distance-to-customer vs cost

The mechanism. Amazon positions inventory close to forecasted demand so that the outbound leg — the expensive, promise-bound leg — is short **[well-known public Amazon practice]**. Inbound placement is the upstream half of that decision: which node receives the units, given regional demand, node capacity and inbound cost. The trade is never "cheapest inbound" — it is distance-weighted delivered cost under a capacity constraint, with the customer promise as the boundary condition. Speed and cost are one table, not two arguments.

Network safety-stock positioning. Because pooling variability upstream beats duplicating buffers everywhere **[industry standard]**, the cheap place to hold uncertainty is upstream (AWD-style bulk) and the fast place is forward (FC pickface); the worked question is how much buffer each layer carries (INV-B) and when a transfer beats a stockout (INV-D).

Weeks of cover, health, and the FBA seller view

CONCEPT	DEFINITION	OPERATOR'S ACTION RULE	LABEL
Weeks of cover (WOC)	WOC = on-hand ÷ weekly forecast	< 2 → replenish / expedite · 2–4 → healthy · 4–6 → watch · > 6 → excess action (worked in INV-A).	**[industry standard]**
Excess / aged	Units above the cover ceiling, and units aging against storage-fee cliffs.	Price, transfer, or remove — in that order of preference.	**[industry standard]**
Stranded	Physically present, not sellable (listing, hold, or data break) — inbound's silent inventory.	Fix the listing/hold at the source; a stow without a sellable record is not availability.	**[well-known public Amazon practice]**
In-stock rate	Share of demand-facing selection available to promise.	The north-star inventory output; every inbound metric is an input to it.	**[industry standard]**
IPI (FBA seller view)	Inventory Performance Index components: excess inventory %, sell-through, stranded inventory, in-stock rate.	Sellers manage to the same four levers; restock limits make network capacity legible to sellers **[well-known public Amazon practice]**.	**[well-known public Amazon practice]**
Transfers / rebalancing	FC↔FC moves that correct placement error.	Run them against the breakeven (INV-D), never as a reflex.	**[well-known public Amazon practice]**

INV-A · WEEKS OF COVER — THREE SKUS, THREE ACTIONS

STEP	SKU	FORMULA → SUBSTITUTION	WOC	ACTION (RULE FROM THE TABLE ABOVE)
1	A	`4,200 ÷ 600/wk = 7.0`	7.0	> 6 → excess action: to reach 4.0 WOC move 4,200 − 4.0×600 = **1,800 units** (transfer feeds INV-D)
2	B	`1,350 ÷ 450/wk = 3.0`	3.0	2–4 → healthy; no action, watch variance
3	C	`500 ÷ 500/wk = 1.0`	1.0	< 2 → expedite: stockout in 500 ÷ (500÷7) = **7.0 days** absent receipt; pull the inbound appointment forward or re-source

The sentence to say: "WOC turns a warehouse full of units into three sentences: A is seven weeks of stock and I'm moving 1,800 units; B is fine; C is a stockout next Friday unless the appointment moves."

INV-B · SAFETY STOCK — Z·Σ·√LT, SUBSTITUTED, AND THE TWO LEVERS

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Demand and variability	`d = 500/day; σ_d = 120/day`	illustrative SKU at one FC	—
2	Demand over lead time	`500 × 7 = 3,500`	lead time LT = 7 days (illustrative)	3,500 units cycle exposure
3	σ over lead time	`σ_LT = 120 × √7 = 120 × 2.6458 = 317.5`	independent daily demand [industry standard]	317.5 units
4	Safety stock at 95%	`SS = 1.645 × 317.5 = 522`	z(95%) = 1.645 [industry standard]	**522 units = 522 ÷ 500 = 1.04 days**
5	Service lever: 99%	`SS = 2.326 × 317.5 = 739; 739 − 522 = +217`	z(99%) = 2.326	+217 units = **+41.6% buffer for +4 pts**
6	Lead-time lever: 7 → 5 days	`SS = 1.645 × 120 × √5 = 1.645 × 268.3 = 441; 522 − 441 = −81`	√5 = 2.2361; 120 × 2.2361 = 268.3	−81 units = **−15.5% buffer, free**

The sentence to say: "Service points are bought at an accelerating price — 95 to 99 costs forty percent more buffer — but lead time is the free lever: two days off the lane releases fifteen percent of the buffer with the service level untouched. That's why measured lead times, not brochure ones, are an inventory program" (this is exactly the S8 story — §5).

INV-C · PLACEMENT MINI-MODEL — 2 FCS × 3 DEMAND REGIONS, CAPACITY-CONSTRAINED

Distance-weighted delivered cost per unit **[illustrative]**. Demand: R1 = 2,000/wk, R2 = 3,000/wk, R3 = 1,000/wk (6,000 total). FC-West capacity 2,500 units/wk; FC-East 4,000.

COST $/UNIT	R1 (2,000)	R2 (3,000)	R3 (1,000)
FC-East	$1.20	$2.40	$3.60
FC-West	$3.40	$2.10	$1.30

STEP	DECISION	FORMULA → SUBSTITUTION	WEEKLY COST
1	Unconstrained optimum: R1→East, R2→West, R3→West	2,000×1.20 + 3,000×2.10 + 1,000×1.30 = 2,400 + 6,300 + 1,300	$10,000
2	West capacity binds at 2,500: give West its cheapest regions first — R3 (1,000), then R2 partial (1,500)	1,000×1.30 + 1,500×2.10 = 1,300 + 3,150	West serves 2,500
3	East takes R1 + residual R2 (1,500); load check 2,000 + 1,500 = 3,500 ≤ 4,000	2,000×1.20 + 1,500×2.40 = 2,400 + 3,600	East serves 3,500 ✓
4	Constrained total	2,400 + 1,300 + 3,150 + 3,600	**$10,450**
5	Capacity premium	10,450 − 10,000 = 450; 450 ÷ 10,000 = 4.5%	**$450/wk (+4.5%)**
6	Shadow price check	(2.40 − 2.10) × 1,500 = 0.30 × 1,500 = 450	each added West slot is worth $0.30 on R2 — the model agrees with itself

The sentence to say: "Placement is a constrained assignment, not a map. Here the constraint costs four and a half points, and the shadow price — thirty cents a unit — is the number I'd take to the capacity team to argue for one more West slot, and to the finance review to explain the premium."

INV-D · TRANSFER BREAKEVEN — WHEN REBALANCING BEATS A STOCKOUT

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Transfer cost per unit	$1.10 handling + $1.90 freight = $3.00	illustrative FC↔FC economics	$3.00/unit
2	Expected stockout cost per unit	$12 × 0.40 = $4.80	lost margin $12/unit; P(stockout before inbound lands) = 40% (illustrative)	$4.80/unit
3	Decision rule	3.00 < 4.80 → transfer	transfer when cost < expected loss	**transfer wins by $1.80/unit**
4	Breakeven probability	p* = 3.00 ÷ 12 = 0.25	transfer when stockout probability > 25%	p* = 25%
5	At scale (SKU A's 1,800 units)	spend 1,800 × 3.00 = 5,400; expected loss 1,800 × 4.80 = 8,640; net 8,640 − 5,400 = 3,240	the 1,800 units from INV-A	**+$3,240 net — and the premium is priced, not argued**

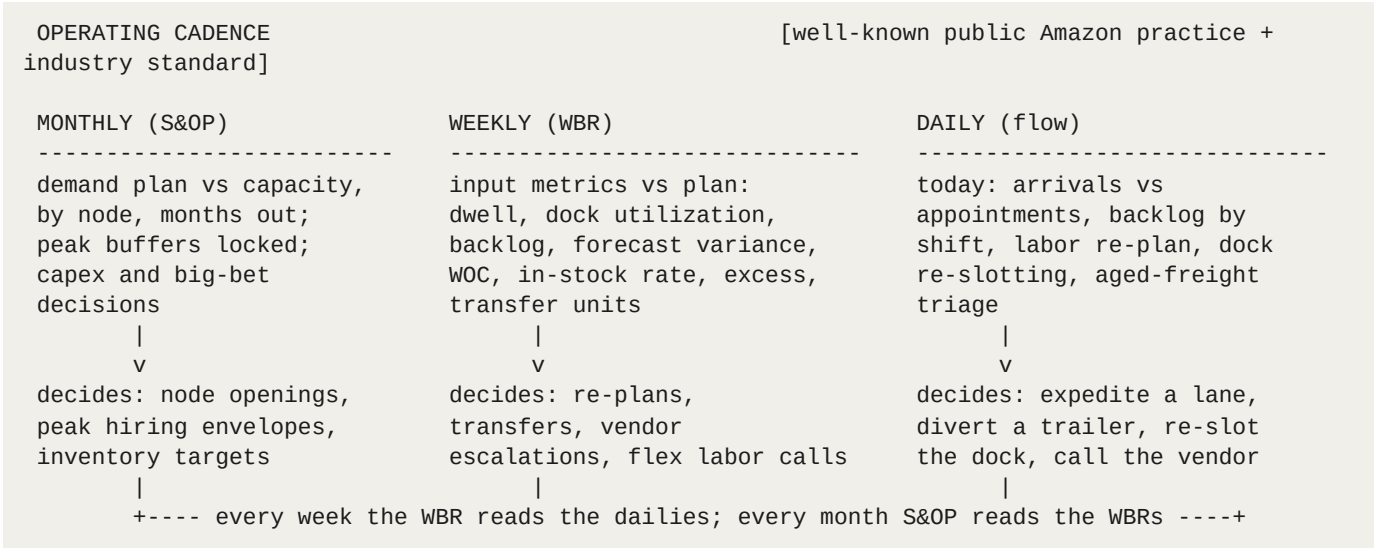
The sentence to say: "Transfers are insurance with a deductible. At a twelve-dollar margin the breakeven is a twenty-five percent stockout probability — below that I let it ride, above it I move the freight, and either way the decision is arithmetic, not adrenaline."

INV-E · VENDOR OTIF / FILL-RATE → THE IN-STOCK CHAIN

STEP	LINK	FORMULA → SUBSTITUTION	VALUE
1	Planned PO volume	`10,000 units/wk = 1,429/day demand`	10,000 units (illustrative)
2	× fill rate	`10,000 × 0.92 = 9,200`	9,200 shipped (800 short)
3	× on-time	`9,200 × 0.88 = 8,096`	8,096 OTIF units = **81.0% of plan**
4	Weekly deficit	`10,000 − 8,096 = 1,904`	1,904 units (19.0% of plan)
5	Safety-stock absorption	`522 ÷ 1,429 = 0.37 days; residual 1,904 − 522 = 1,382`	the INV-B buffer covers about a third of a day — it was sized for demand noise, not vendor failure
6	In-stock proxy for the week	`(10,000 − 1,382) ÷ 10,000 = 86.2%`	**~86% in-stock absent recovery — a 14-point hit from one vendor's 92/88 week**

The sentence to say: "OTIF isn't a vendor scorecard, it's the first link of the in-stock chain: fill times on-time is eighty-one percent of plan, and a cycle buffer sized for demand noise absorbs a third of a day of it. So vendor management is inventory management — measured lead times, escalation thresholds, and a recovery path that starts at the ASN, not at the stockout."

4 · Operating cadence — one drumbeat, three tempos



Peak (Q4) planning rhythm [industry standard]

WHEN	WHAT GETS DECIDED	THE INBOUND-SPECIFIC WORK
Q2–early Q3	Network peak plan: capacity envelopes by node, hiring envelopes, inventory targets.	p50/p90 forecast distribution locked (CAP-E); cube plan vs effective capacity (CAP-C); vendor appointment and freight-plan commitments.
Late Q3	Readiness: flex labor pipelines, overflow/storage contingency, freeze windows.	Learning-curve staffing math paid early (CAP-B step 5–6); appointment smoothing rules published to vendors/carriers.
Q4 in-peak	Execution: daily flow cadence dominates; WBR shifts to peak inputs.	Dwell and backlog watched hourly (CAP-D); transfers only against the breakeven (INV-D); escalations on same-sign forecast variance.
January	Debrief: what the plan got wrong, written down while it's fresh.	Variance autopsy by node; buffers released on evidence, not relief (the S6/S8 release rule, §5).

Escalation norms. Escalate with the arithmetic attached: the metric, its frozen definition, the variance, the root cause so far, and the decision you need [industry standard]. Same-sign variance three periods running is a plan problem and goes up as a re-plan request, not a floor problem and goes down as pressure. A miss you report with a recovery path is a WBR input; a miss discovered by someone else is a trust problem — the S11 discipline (§5).

5 · Tie-to-your-stories — the master packet, spoken in inbound

Each story maps to one inbound concept; the third column is the sentence that makes the mapping out loud. Canon numbers carry their basis labels; stories stay allocated to their panel owners — here they are *vocabulary*, not re-told STARs.

STORY (CANON)	INBOUND CONCEPT	THE SENTENCE THAT TIES THEM
S8 — Elementum measured lanes (200 lanes; promise at p99; 99% lead-time accuracy; 580K units / ~$1.5M buffer released, record, one-time)	Inbound lead-time reliability	"Every inbound plan and every safety-stock formula is only as honest as its lead times — I replaced brochure lead times with measured distributions and released buffer on evidence; INV-B's free lever is that story."
S6 — Elementum supply/demand view (forecast accuracy 70% → 85%; $2M safety stock released under the both-errors rule, record, one-time)	Network safety-stock positioning	"Buffer gets released where forecast error AND lead-time error both improved — a signed release rule, not a victory lap; that's how I'd run NACF's inventory health reviews."
S2 — Sephora UDC comingle (two warehouses → one pool; +15% efficiency, signed; robotics behind a published 98% gate)	Capacity pooling	"Pooling variance beats adding capacity: one pool removed duplicate handling before anyone asked for a bigger building — the same logic as holding cube slack upstream instead of over-building the FC."
S13 — Manhattan Cyber Monday (3 DCs / 100 stores; 99% vs 95% fill; ~$8M protected vs ~$6M at risk, record pair)	Peak capacity under failure	"Peak is when the CAP-D nonlinearity stops being theory — I rebalanced by promise-at-risk at 4 AM, and the post-mortem shipped the heartbeat and the playbook so the next peak ran boring."
S18 — Sephora dual-DC WMS migration (0.5% tolerance published with Finance; mocks 7% → 1.8% → 0.3%; rehearsed fallback for ~100 of ~400 RF devices)	Launch / cutover readiness	"New capacity or a new process goes live behind a published tolerance and a rehearsed fallback — exactly how I'd bring a new IXD lane or a peak process into the network."
S19 — Elementum promise dial (k = 1.645σ one governed dial; fast-promise share 7.6% → 39.6% at a held 95% floor — all model, labeled)	Promise from inventory position	"The date the customer sees is priced from inventory position and lane variability — placement (INV-C) sets the position, the dial prices the promise."

STORY (CANON)	INBOUND CONCEPT	THE SENTENCE THAT TIES THEM
S11 / S12 — Flexport visibility & roadmap (boarding behind published gates; CTS $0.66 → $0.03 across 7 → 68 customers; month-18 plan → value in month 6)	Upstream visibility into inbound	"Inbound starts before the truck: ASN-grade visibility and graduation gates upstream are what let a dock plan survive contact with real vendors — and re-sequencing on cost-to-serve is how I'd order the roadmap to get there."
S9 — Elementum lane onboarding (597 lanes = 80% of spend live in month one; ≥95% valid-ASN gate held twice; completeness 60% → 95%)	ASN / inbound data quality	"Bad ASNs manufacture surprise at the dock — a gate on data quality upstream is capacity protection downstream."

> **Discipline in the room.** These are one-sentence bridges, not re-tellings — the full STARs belong to their allocated interviewers per the panel packets. Never mix the loop packet's conflicting numbers into these bridges; never present any modeled dollar as record.

6 · Per-interviewer questions — what they'll ask, and what to ask them

For each interviewer: six likely NACF-inbound domain questions with a one-line answer strategy (pointing at a booklet section or a canon story), then four questions for you to ask them. Answer strategies are one breath long — the booklet section carries the depth.

Snigdha Tripathy — Sr Manager, Supply Chain · HIRING MANAGER

> **Lens.** Placement and IXD are her home turf (EU IXD leader; built the team ground-up; 50+ FC launches). She judges fit for the actual work: network process design, value made legible, and judgment under unit economics. Talk placement slowly and precisely.

#	LIKELY NACF-INBOUND QUESTION	ONE-LINE ANSWER STRATEGY → POINTER
1	**Walk me through how you'd decide where inbound inventory lands across our network.**	Constrained assignment on distance-weighted cost, capacity binds, shadow price names the ask — INV-C, said slowly; bridge to S8 measured lanes.
2	**How do you know the network has enough inbound capacity for Q4?**	Forecast → receive plan → doors at the 85% ceiling → UPH labor → p50/p90 flex — CAP-A/B/E as one spoken chain; S13 for in-peak failure.
3	**Tell me about a time you built a process or capability from nothing.**	S11: boarding process behind published gates + weekly input/output review — mirrors her ground-up IXD build; gates published before wave 1.
4	**A strategic vendor misses OTIF three weeks running into peak. What do you do?**	Quantify the in-stock chain first (INV-E), then escalate with the arithmetic — measured lead times, recovery path, and a re-plan, not blame (S8).
5	**How do you run your weekly business review — inputs, outputs, decisions?**	§4 cadence + S11 mechanics: frozen definitions signed by owners, inputs tagged to outputs, reds pre-decomposed with dated fixes.
6	**How do you trade speed against cost in placement without losing the customer promise?**	Promise as the boundary condition, then one priced table — INV-C premium (+4.5%) and INV-D breakeven; S19's governed dial as the promise-side analogue.

Questions to ask her

- "What's the charter for this role in its first year — which inbound outcomes does it own end-to-end, and which does it influence?"
- "Where does inbound hurt most in NACF right now — dock capacity, labor, cube, or placement accuracy — and what would you want moved first?"
- "How is this seat measured at your WBR — which two or three input metrics would tell you, six months in, that you hired right?"
- "You built placement and IXD capability ground-up in EU — what did that teach you about sequencing people, process and mechanisms that I should know walking in?"

Ryan Hoerger — Manager, Placement / Transfer Analytics

> **Lens.** Ex-Sr BIE, Booth MBA, Siebel Scholar, Nike analytics — statistical rigor is the love language. Show the distribution behind the number, name your assumptions, and let the math be audited.

#	LIKELY NACF-INBOUND QUESTION	ONE-LINE ANSWER STRATEGY → POINTER
1	**How would you model optimal inventory placement across FCs?**	INV-C cost matrix → capacity-constrained assignment → shadow price; then validate against actuals and re-solve on a cadence.
2	**When is a transfer worth doing?**	INV-D: transfer cost vs lost margin × stockout probability; breakeven p* = 25% on illustrative economics — a rule, not a reflex.
3	**How do you size and position safety stock across a network?**	INV-B $z \cdot \sigma \cdot \sqrt{LT}$ with both levers (service points cost +41.6%, lead time frees −15.5%); pool upstream, release on evidence (S6 both-errors rule).
4	**How do you measure forecast quality, and what do you do with the variance?**	§2 metrics: same-sign variance three weeks running is bias — re-plan; S6 70 → 85% accuracy program as the earned credential.
5	**Do you size buffers to p50 or p90?**	CAP-E: fixed to ~p50–p70, flex the tail — show the 23%-idle vs 1-day-backlog arithmetic, then the ≈14-head flex answer.
6	**How do you keep a model from going stale or getting gamed?**	Drift flags + refresh cadence (S8), frozen definitions signed by owners (S11), honest denominators — a truncated week is a data miss, not a result.

Questions to ask him

- "How are the placement and transfer models measured — what's the scoreboard, and how often do operators override the recommendation?"
- "Where does the model disagree with operators most often, and who's usually right?"
- "How fresh is the demand and lead-time signal the placement model sees — and what would you measure if you could see it fresher?"
- "What analysis does your team wish it had that doesn't exist today — and what's kept it from existing?"

Rocky Goforth — Sr Manager, Infrastructure

> **Lens.** Infrastructure owns the physical envelope — buildings, docks, automation, peak readiness. Speak capacity as physics: constraints, nonlinearities, launch discipline.

#	LIKELY NACF-INBOUND QUESTION	ONE-LINE ANSWER STRATEGY → POINTER
1	**What actually constrains an FC's inbound throughput — and which one binds first at peak?**	§2 three currencies; dock slots bind first (least elastic), labor second (slow to hire), cube last-then-suddenly — CAP-A/C framing.
2	**Why does dwell blow up when docks run hot?**	CAP-D: ρ/(1−ρ) nonlinearity — +10 points of utilization ≈ 3× dwell; that's why the plan holds dock slack like safety stock.
3	**How should we be planning infrastructure for Q4?**	§4 peak rhythm + CAP-E p50/p90; in-peak failure leadership from S13 (promise-ranked rebalancing, post-mortem that shipped mechanisms).
4	**Tell me about bringing automation into a live building.**	S2: Locus robotics behind a published 98% gate with a pre-built throttle — week two hit 94, throttled to ~30%, root-caused, full volume week three.
5	**How do you launch new capacity or process without wobbling the network?**	S18: published 0.5% tolerance with Finance, three mocks 7 → 1.8 → 0.3%, rehearsed fallback when ~100 of ~400 RF devices failed.
6	**When capacity and cost fight, how do you decide?**	Price the constraint — INV-C shadow price ($0.30/unit) turns "we need more doors" into a dollarized ask; fixed-to-p50 / flex-the-tail (CAP-E).

Questions to ask him

- "Which capacity constraint is binding hardest in NACF going into this peak — dock, labor, or cube?"
- "How far ahead are capacity decisions locked, and what's the escalation path when demand beats the envelope?"
- "What should Supply Chain bring to Infrastructure earlier than it does today?"
- "Where has a new building or an automation ramp surprised you recently — and what did the surprise teach the planning process?"

Zhaowen Lan — Sr Manager, Supply Chain · peer

> **Lens.** A peer who shares the cadence. He is listening for how you run mechanisms other people own — the WBR, the handoffs, the disagreement that stays professional.

#	LIKELY NACF-INBOUND QUESTION	ONE-LINE ANSWER STRATEGY → POINTER
1	**How do you run a WBR that people actually trust?**	S11 mechanics: 5–7 frozen input metrics, each signed by its owner, tagged to the output it moves, reds pre-decomposed with dated fixes.
2	**An input metric owned by another team is red every week. What do you do?**	Pre-decompose with the owner, bring the dated fix; if the definition is wrong, fix the definition in public — never the number in private (S11's discarded 99.4% week).
3	**How do you manage the S&OP-to-WBR-to-daily handoff?**	§4 cadence diagram: S&OP locks envelopes, WBR reads variance, daily flow re-plans; same-sign variance three weeks running escalates as a re-plan request.
4	**Tell me about a disagreement with a partner team and how it resolved.**	S12: privately first with the numbers, reversal trigger written before the swing, the plan's owner ratifying the flip in the room.
5	**How do you keep inventory healthy across the network — excess vs in-stock?**	INV-A WOC action rules + §3 health definitions (excess/aged/stranded); in-stock as the north-star output, everything else an input.
6	**What would your first 90 days look like in this seat?**	Book of record and primary metric first → stocks-and-flows map → the broken invariant → one input/output review running by week 6 — §4 plus master-packet §3.0 discipline.

Questions to ask him

- "What does your team own end-to-end, and where are the seams with this role — the places two people could both think the other has it?"
- "What does a great week look like at your WBR — and which input is the last to turn green?"
- "How do the placement, capacity, and inventory-health reviews nest into each other here?"
- "What's the most common way a new Senior SCM gets surprised in their first quarter?"

Venkata Josyula — Principal PM, FBA · bar raiser

> **Lens.** Principal PM for FBA plus bar-raiser duties: the seller is a customer, LP answers must be clean STARs with honest numbers, and every mechanism should end at a customer outcome.

#	LIKELY NACF-INBOUND QUESTION	ONE-LINE ANSWER STRATEGY → POINTER
1	**How does inbound look from the seller's side of FBA?**	§3 seller view: ship-to-Amazon → appointment → check-in → available; IPI's four components (excess, sell-through, stranded, in-stock) and restock limits as network capacity made legible **[well-known public Amazon practice]**.
2	**Why Amazon — and why this role?**	The master's three reasons — the forecast (Elementum), the weekly review (Flexport), one version of the truth (Sephora) — landing on inbound as the seam where all three meet.
3	**Tell me about a time you failed.**	S18's RF miss: scored the task by effort when impact was what mattered; ~100 of ~400 devices, rehearsed fallback fired, lesson — single-point dependencies get a what-if plan regardless of size.
4	**Seller experience and network cost pull apart — how do you trade them?**	Price the trade, publish it, govern it: INV-D breakeven as priced risk; S19's one-dial discipline — the no goes in the comparison table as a priced row, not a meeting minute.
5	**How would you improve FBA inbound?**	Upstream first: visibility before the truck (S11/S9 ASN quality), appointment reliability, check-in-to-available time (§1 time-to-available); AWD as the cheap-cube shock absorber **[well-known public Amazon practice]**.
6	**Tell me about a time you chose long-term over short-term pressure.**	S12: an 18-month committed roadmap re-sequenced to value in month 6 at 66% less effort — with the reversal trigger written before the swing and operators' delay priced and published.

Questions to ask him

- "Where does the FBA inbound experience break seller trust most — appointments, check-in-to-available time, or restock limits?"
- "How is AWD changing the conversation with sellers about holding inventory upstream?"
- "What makes a Supply Chain Manager a great partner to a Principal PM — and what makes one hard to work with?"
- "Which seller-facing inbound metric would you most want this team to move this year?"

7 · Math appendix — every formula, symbolic → substituted → computed

The night-before page. Each row: the formula in symbols, the substitution from the booklet's worked example, the computed result, and where it lives. All inputs **[illustrative]** unless labeled otherwise; recompute each row from the row above it.

#	FORMULA (SYMBOLIC)	SUBSTITUTION	COMPUTED	SECTION
A1	`trailers/day = V ÷ u_c ÷ c_p ÷ p_t`	`120,000 ÷ 6 ÷ 40 ÷ 26`	`= 19.2 trailers/day`	CAP-A
A2	`doors = (trailers/day ÷ hours) ÷ unload_rate ÷ ρ_max`	`(19.2 ÷ 10) ÷ 0.5 ÷ 0.85`	`= 4.5 → 5 doors`	CAP-A
A3	`direct hours = V ÷ UPH; paid = ×(1+indirect)`	`120,000 ÷ 300 = 400; ×1.15`	`= 460 paid h; ÷10 = 46 heads`	CAP-B
A4	`blended UPH = (1−n)·UPH + n·UPH·LC`	`0.7×300 + 0.3×300×0.6 = 264`	`120,000 ÷ 264 = 454.5 h (+13.6%)`	CAP-B
A5	`util_true = I·cube ÷ (C × 0.85)`	`2.0M×0.25 ÷ (1.2M×0.85)`	`= 500,000 ÷ 1,020,000 = 49.0%`	CAP-C
A6	`queue factor = ρ ÷ (1−ρ); dwell ≈ service ÷ (1−ρ)`	`ρ=0.95: 19.0; 2.0 ÷ 0.05`	`= 40 h dwell (vs 13.3 h at 85%)`	CAP-D
A7	`flex heads = (p90 − p50) ÷ UPH_blend ÷ shift`	`36,000 ÷ 264 ÷ 10`	`= 13.6 ≈ 14 flex associates`	CAP-E
B1	`WOC = on-hand ÷ weekly forecast`	`A: 4,200÷600; B: 1,350÷450; C: 500÷500`	`= 7.0 / 3.0 / 1.0 weeks`	INV-A
B2	`excess units = on-hand − WOC_target × forecast`	`4,200 − 4.0×600`	`= 1,800 units to move`	INV-A
B3	`SS = z × σ_d × √LT`	`1.645 × 120 × √7 = 1.645 × 317.5`	`= 522 units = 1.04 days`	INV-B
B4	`SS(99) − SS(95) = (z99 − z95)·σ_LT`	`(2.326 − 1.645) × 317.5`	`= +217 units (+41.6%)`	INV-B
B5	`SS(LT=5) = z × σ_d × √5`	`1.645 × 120 × 2.2361`	`= 441 units (−81; −15.5%)`	INV-B
B6	`cost = Σ d_r × c_{r,f} s.t. node capacity`	`2,000×1.20 + 1,000×1.30 + 1,500×2.10 + 1,500×2.40`	`= $10,450/wk (vs $10,000 unconstrained)`	INV-C
B7	`shadow price = (c_far − c_near) × units_displaced`	`(2.40 − 2.10) × 1,500`	`= $450/wk = the capacity premium`	INV-C
B8	`transfer if cost_t < margin × P(stockout); p* = cost_t ÷ margin`	`3.00 < 12 × 0.40 = 4.80; p* = 3.00 ÷ 12`	`= transfer; p* = 25%; net +$3,240 on 1,800 units`	INV-D
B9	`OTIF_units = PO × fill × on_time`	`10,000 × 0.92 × 0.88`	`= 8,096 units = 81.0% of plan`	INV-E
B10	`in-stock proxy = 1 − (deficit − SS) ÷ PO`	`1 − (1,904 − 522) ÷ 10,000`	`= 86.2% weekly in-stock proxy`	INV-E

Recompute discipline: A1→A2 share 19.2; A3→A4 share the 120,000 and 300 UPH; B3→B4→B5 share σ_d = 120 and z values; B2's 1,800 feeds B8's scale row; B9's deficit feeds B10. If any row doesn't reproduce from its own substitution, the row is wrong — say the formula, not the memory.